ACADEMIUM ENGINE PACKET

Section 5: Causal Explanation Generation

Version: 2.0 **Status:** Active Specification **Scope:** Applies to every KPI movement displayed to the student, across every turn, in every case. No KPI movement may be displayed without a causal explanation generated and ready for delivery. **Runs after:** Section 4 (KPI Computation — provides the movement set), Section 3 (Consequence Generation — provides the narrative context), Section 2 (Signal Extraction — provides the student's specific reasoning signals) **Outputs sent to:** Section 6 (Response Assembly — receives the complete causal explanation set, one per displayed KPI movement)

5.1 Purpose and Pedagogical Foundation

The causal explanation is the engine's answer to the most important question a student can ask after seeing a metric move: **"Why did that happen?"**

It is not a summary of the consequence narrative. It is not a restatement of the KPI movement. It is a precise, two-to-four-sentence causal chain that connects the student's specific decision to the organizational effect measured by that KPI — explaining the mechanism, not just the direction.

Without this, metric movements feel arbitrary. A student who sees Team Morale drop and cannot trace why will either feel punished without cause or dismiss the simulation as unreliable. Either outcome destroys trust. The causal explanation is the evidence that the simulation is behaving logically, consistently, and in direct response to what the student actually decided.

The Distinction Between Narrative and Explanation

Section 3 produces a consequence narrative — a scene. It shows what happened in the organization: who reacted, what shifted, what emerged. Section 5 produces a causal explanation — a mechanism. It isolates a single chain of cause and effect and makes it legible.

A student reading the narrative experiences the simulation. A student reading the causal explanation understands it. Both are necessary. Neither replaces the other.

Research Basis

Feedback Intervention Theory (Kluger & DeNisi, 1996): The most effective feedback is specific, task-directed, and causally grounded. Generic feedback ("your score dropped") produces no learning. Mechanistic feedback ("this specific action produced this specific

organizational effect through this causal pathway") produces durable understanding. The causal explanation is the operational implementation of this principle.

Cognitive Load Theory (Sweller, 1988): The "Why?" affordance is student-activated — the student chooses when to read the explanation. This is deliberate. Displaying the explanation automatically alongside the narrative and metric would create information overload. Student control over when to access the explanation reduces extraneous cognitive load and respects the student's processing pace.

Constructivism (Piaget, 1971; Vygotsky, 1978): Before reading the explanation, the student has already experienced the consequence and observed the metric movement. The causal explanation arrives after they have had the chance to form their own hypothesis about what caused the change. This sequencing — experience first, explanation accessible second — supports active knowledge construction rather than passive information reception.

Metacognition (Flavell, 1979): Causal explanations are metacognitive scaffolds. By making the causal chain explicit, the engine helps students build more accurate mental models of organizational dynamics. Over multiple turns, students who engage with causal explanations develop better predictive reasoning — they begin anticipating consequences before seeing them.

5.2 Input Requirements

The causal explanation generator must have access to all of the following before generating any explanation. These inputs ensure that every explanation is specific, accurate, and anchored to what the student actually did.

Input	Source	Purpose
KPI movement set (direction, tier, KPI ID)	Section 4	Defines which explanations need to be generated
Signal extraction data (per KPI domain)	Section 2	Enables signal anchoring at Engaged and Integrated bands
RDS band	Section 2	Governs explanation depth and length
Student's original input text	Input Reception	Enables direct or paraphrased reference to student's reasoning
Consequence narrative	Section 3	Ensures explanation and narrative are consistent; prevents contradiction
Case directional architecture	Case definition	Provides the organizational logic that makes causal chains defensible
Prior KPI accumulation state	Section 4	Enables accumulated context references when relevant

5.3 The Three-Part Causal Structure

Every causal explanation is built from three components. These components may appear in any order, but all three must be present for Engaged and Integrated band explanations. Surface band explanations may compress the structure but must contain all three implicitly.

Component 1: The Decision Reference

A statement of what the student decided, framed in organizational terms. This is not an evaluation of the decision — it is a factual anchor that connects the explanation to the specific action taken.

Rules:

- Written in third-person organizational framing: *"The decision to..."* or *"Following the commitment to..."* — never *"You decided to..."* or *"Your choice to..."*
- References the specific direction or action, not a general category
- For MCQ turns: references the selected option's strategic posture

- For FR turns: references the extracted intent signal (or the most prominent signal if intent was absent)

Correct: *"The decision to restructure internal communication protocols before confirming any revised timeline to the client..."* **Incorrect:** *"You decided to communicate with your team..."* (second-person; too vague)

Component 2: The Causal Mechanism

An explanation of HOW the decision produced an effect in this specific KPI's domain. This is the most important component — it is what transforms the explanation from a label into a learning moment.

Rules:

- Must identify the organizational pathway, not just the outcome
- Must be specific to the KPI domain — a mechanism for K2 (Team Morale) is different from one for K5 (Stakeholder Trust), even if they stem from the same decision
- Must be causally defensible — the mechanism must reflect realistic organizational dynamics, not arbitrary assignments
- Must NOT evaluate whether the mechanism was good or bad

Correct: *"...signaled to the operations team that leadership was aware of their capacity constraints. This reduced the team's sense of being set up for an unmanageable commitment, which directly affected their confidence and willingness to engage with the project."* **Incorrect:** *"...which was the right approach because morale always improves when communication happens."* (Evaluative; overly generic)

Component 3: The Directional Connection

A brief statement connecting the mechanism to the specific direction of movement (positive or negative). This closes the explanation — it completes the chain from decision to metric direction.

Rules:

- Should be the shortest component — one sentence is sufficient
- Must connect explicitly to the metric that moved (not just the general domain)
- Must not imply a judgment about whether the direction was desirable

Correct: *"As a result, team cohesion indicators moved in a positive direction for this phase of the project."* **Incorrect:** *"This was fortunate, as the team really needed a morale boost at this stage."* (Emotionally loaded; implies the outcome was lucky rather than caused)

5.4 RDS-Governed Explanation Depth

Explanation depth scales to the RDS band received from Section 2. This ensures that explanations are proportionally rich — a student who provided minimal reasoning receives a clear, direct explanation; a student who provided multi-signal reasoning receives an explanation that reflects the complexity of what they engaged with.

Surface Band (RDS 1–4)

Parameter	Specification
Length	2 sentences
Structure	Decision Reference + Directional Connection (mechanism implied, not stated)
Signal anchoring	None — explanation references organizational dynamics generically
Tone	Direct and factual

Pedagogical rationale: Surface band students are at the early stages of engagement. A 2-sentence explanation is clear, accessible, and doesn't overwhelm. The mechanism is implied rather than stated — the student can infer it from the decision reference and direction, which itself is a light metacognitive exercise.

Engaged Band (RDS 5–9)

Parameter	Specification
Length	2–3 sentences
Structure	All three components present
Signal anchoring	Light — references domain of student's PRESENT-level signals
Tone	Factual with organizational specificity

Integrated Band (RDS 10–15)

Parameter	Specification
Length	3–4 sentences

Parameter	Specification
Structure	All three components present with full case-specific grounding
Signal anchoring	Direct — paraphrases or echoes STRONG-level signals from student's response
Tone	Factual, nuanced, reflects organizational interconnections

Pedagogical rationale: Integrated band students demonstrated multi-signal reasoning. Seeing their own reasoning echoed back in the causal explanation — reframed as organizational logic — validates that their thinking was observed and engaged with. This builds self-efficacy and reinforces the cognitive moves they made.

5.5 Signal Anchoring: Making Explanations Responsive

Signal anchoring is the mechanism by which causal explanations at Engaged and Integrated bands reference the student's own reasoning. It is what makes the simulation feel genuinely responsive — not like a predetermined script — and is one of the most pedagogically important features of the engine.

What Signal Anchoring Is

Signal anchoring means that the causal explanation for a KPI movement paraphrases, echoes, or directly references the reasoning the student demonstrated in their input, translated into organizational language.

It does NOT mean:

- Quoting the student's exact words back to them
- Praising the student for their reasoning
- Implying that because they reasoned well, the outcome was better

It MEANS:

- Recognizing what the student was thinking about and reflecting it in the causal mechanism
- Making the student's reasoning the logical antecedent of the organizational effect

Signal Anchoring by Signal Type

Student Signal	Anchoring Approach
Intent (STRONG)	The decision reference in Component 1 directly reflects the student's stated priority: *"The decision to prioritize internal alignment before external commitments..."*
Justification (STRONG)	The causal mechanism in Component 2 references the reasoning the student provided: *"The logic of addressing internal constraints before client-facing commitments reduced the risk of overcommitment..."*
Tradeoff Awareness (STRONG)	The directional connection in Component 3 validates that the cost the student anticipated was the actual organizational mechanism: *"The anticipated impact on delivery capacity materialized as a Tier 1 reduction in operational efficiency indicators..."*
Stakeholder Awareness (STRONG)	The causal mechanism references the specific stakeholder the student named: *"The decision to involve the operations director before confirming client commitments addressed a key coordination gap that had been generating internal friction..."*
Ethical Awareness (STRONG)	The causal mechanism references the ethical dimension the student surfaced: *"The commitment to transparency with the client about operational constraints, rather than managing perceptions through selective disclosure, produced a credibility signal that affected external trust indicators..."*

Signal Anchoring When Signals Are WEAK or ABSENT

For WEAK signals: the explanation references the general domain the student touched without specific attribution. *"The team communication direction..."* rather than *"The decision to coordinate with the operations director before confirming timelines..."*

For ABSENT signals: the explanation uses the case's organizational logic without referencing student reasoning. The explanation is still causally accurate but generic. The student receives correct information about what happened without having their reasoning reflected back — because there was no reasoning to reflect.

5.6 MCQ Turn Explanations

For Turn 1 (MCQ), causal explanations are generated from the pre-authored tradeoff signature, not from extracted student signals. The explanation still follows the three-part structure but draws on the option's strategic posture rather than individual signal content.

MCQ Explanation Rules

- **Decision Reference:** References the strategic posture of the selected option, not the option's label or text
- **Causal Mechanism:** Derived from the pre-authored directional logic of the tradeoff signature
- **Signal Anchoring:** Not applicable for Turn 1 — no FR signals exist yet
- **Length:** 2–3 sentences regardless of option selection (MCQ responses don't have RDS bands)
- **Tone:** Same mentorship posture as all other explanations

Example MCQ Explanation (K5 ↑ Tier 2 from Option B selection): *"The decision to prioritize transparent communication with key stakeholders established a clear organizational posture early in the project. By signaling responsiveness over containment, the approach reduced uncertainty for parties with a direct stake in the outcome, which created an early positive movement in stakeholder confidence indicators."*

5.7 Anti-Pattern Correction Explanations

When Section 4 applies an anti-pattern correction (forcing a Tier 1 movement to satisfy the tradeoff requirement), the causal explanation must still be generated and must still be causally defensible. The explanation must not reveal that the movement was a correction — it must identify a legitimate organizational reason why that KPI domain was affected by the decision.

Rules for Correction Explanations

1. **The correction must only be applied to a KPI with a genuine indirect causal connection to the decision.** The engine cannot force a K3 (Brand Reputation) movement for an internal team communication decision if no external-facing element was involved. The correction target must be organizationally plausible.
2. **The explanation draws on the case's organizational architecture** — the indirect pathway from decision to KPI — rather than on student signals.
3. **The explanation reflects the indirect nature honestly** — phrases like *"as a secondary effect of..."* or *"the allocation of resources toward this priority created downstream pressure on..."* accurately represent that this is a second-order effect, not a direct outcome.
4. **Never explain the engine's correction logic.** The explanation describes an organizational reality, not a computational process.

5.8 Prohibited Content

All prohibitions from Section 3.7 apply here. The following additional prohibitions are specific to causal explanations:

Prohibited	Why
"This happened because you were [adjective]" (e.g., "too cautious," "not strategic enough")	Psychological evaluation of the student
"If you had done X differently, this would not have happened"	Counterfactual judgment; retroactive answer-giving
"This is because the correct approach was X"	Direct answer-giving through explanation
"Your reasoning led directly to this outcome"	Implies a causal link between reasoning quality and metric direction — the engine does not grade reasoning
"Well done for noticing X"	Praise; evaluative
"This is a common mistake"	Implies a correct path existed; also potentially shaming
Forward-looking directive language	*"To prevent this next time..."* — explanations look backward, never forward

The Hint Test for Explanations

Every causal explanation must pass the Hint Test from Section 3.9 before delivery: does anything in this explanation make one particular response to the next decision prompt seem more clearly correct than others?

The Hint Test is especially critical for causal explanations because they are mechanistic — they explain exactly how organizational cause and effect works. A clear explanation of why K2 dropped ("internal alignment was missing before external commitments were made") is almost automatically a hint about what to do next ("establish internal alignment first"). This tension cannot be fully eliminated, but it can be managed:

Acceptable: An explanation that describes a general organizational mechanism without prescribing a specific corrective action. **Prohibited:** An explanation that names the specific gap or corrective action required.

Too Hint-Like (Prohibited)	Acceptable
"Morale dropped because the team was not consulted before the client commitment was made. Consulting the team first would have prevented this."	*"The sequence of external commitment before internal alignment created uncertainty within the team about whether the commitment was achievable, which affected cohesion indicators."*
"Stakeholder trust declined because transparency was delayed. Earlier disclosure would have maintained trust."	*"The gap between the decision timeline and stakeholder communication introduced ambiguity for parties who had made commitments based on prior information, affecting confidence indicators."*

5.9 The "Why Did This Change?" Affordance

The causal explanation is delivered through a student-activated affordance — a visible, accessible button attached to each KPI movement display. The student decides when and whether to read it.

Affordance Rules

- **Always visible.** The "Why did this change?" button is present for every displayed KPI movement, every turn, without exception. It is never hidden, disabled, or conditional.
- **Student-activated.** The explanation does not auto-display. The student taps or clicks to access it. This preserves the student's metacognitive agency — they can try to reason through the cause themselves before reading the explanation.
- **Persistent.** The student can access the explanation multiple times, including when reviewing prior turns in read-only mode.
- **Immutable.** Once generated, the explanation does not change. The student reads the same content regardless of how many times they access it or what decisions they make subsequently.
- **Not dismissible permanently.** The affordance remains accessible throughout the session and in the final review. Students are not penalized or blocked for not reading explanations.

What the Affordance Displays

The affordance displays:

1. The KPI name (contextual label, per Section 4.9.1)

2. The direction and tier of movement (arrow + magnitude label)
3. The causal explanation text (2–4 sentences)

The affordance does NOT display:

- The RDS band
- The signal scores
- Any competency evidence data
- Any reference to how other students handled the same decision

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5.10 Quality Gates

Before any causal explanation is passed to Section 6 (Response Assembly), it must pass all six quality gates.

Gate	Check	Pass Condition
G1: Decision Reference Present	Does the explanation identify what the student decided in organizational terms?	Yes — specific to this turn's decision
G2: Causal Mechanism Present	Does the explanation explain HOW the decision produced the effect, not just that it did?	Yes — pathway identified, not just direction stated
G3: Prohibited Language	Does the explanation contain any prohibited terms from Section 3.7 or Section 5.8?	Zero prohibited terms
G4: Hint Test	Does the explanation prescribe a specific corrective action for the next decision?	No forward-looking prescriptions
G5: Length Compliance	Is the explanation within the RDS-band-appropriate length (2–4 sentences)?	Yes
G6: Narrative Consistency	Is the explanation consistent with the Section 3 consequence narrative? Does it contradict any established fact or stakeholder position?	No contradictions

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5.11 Worked Examples

All examples in Spanish (Case 001). English translations in brackets.

EXAMPLE A — Surface Band Explanation, K2 (Team Morale) ↑ Tier 1

Student input: *"Hablaría con el equipo."* (RDS = 3, Surface Band) **Signal present:** Intent only (PRESENT) **No signal anchoring applicable**

Generated explanation:

"La decisión de establecer comunicación directa con el equipo interno redujo la incertidumbre sobre cómo el equipo abordaría los compromisos pendientes. Esto generó un movimiento positivo leve en los indicadores de cohesión del equipo."

["The decision to establish direct communication with the internal team reduced uncertainty about how the team would approach pending commitments. This produced a slight positive movement in team cohesion indicators."]

Gate checks: G1 ✓ (decision referenced) G2 — mechanism implied but compressed (2-sentence Surface band) ✓ G3 ✓ G4 ✓ (no forward direction) G5 ✓ (2 sentences) G6 ✓

EXAMPLE B — Engaged Band Explanation, K4 (Operational Efficiency) ↓ Tier 1

Student input: Multi-sentence response. Tradeoff Awareness signal: PRESENT ("esto podría retrasar entregas internas"). **RDS = 7, Engaged Band Light signal anchoring on tradeoff domain**

Generated explanation:

"La decisión de redirigir recursos del equipo hacia la gestión de la relación con el cliente requirió reasignar tiempo y capacidad que estaban comprometidos con procesos internos. La tensión entre los compromisos externos e internos —reconocida en la decisión— se materializó como una reducción en la capacidad de entrega disponible para los proyectos activos. Esto produjo un movimiento negativo leve en los indicadores de eficiencia operativa."

["The decision to redirect team resources toward client relationship management required reallocating time and capacity that had been committed to internal processes. The tension between external and internal commitments — acknowledged in the decision — materialized as a reduction in available delivery capacity for active projects. This produced a slight negative movement in operational efficiency indicators."]

Gate checks: G1 ✓ G2 ✓ (mechanism: resource reallocation → capacity reduction) G3 ✓ G4 ✓ G5 ✓ (3 sentences) G6 ✓ **Signal anchoring note:** *"La tensión entre los compromisos externos e internos — reconocida en la decisión"* echoes the student's tradeoff signal without quoting them directly.

EXAMPLE C — Integrated Band Explanation, K5 (Stakeholder Trust) ↑ Tier 2

Student input: All five signals at STRONG/PRESENT, including: Stakeholder Awareness (named the operations director), Ethical Awareness (transparency with client), Intent (renegotiate timeline before confirming). RDS = 14, Integrated Band.

Generated explanation:

"La decisión de involucrar a la directora de operaciones antes de confirmar cualquier plazo al cliente estableció un proceso de validación interna que protegió la credibilidad de los compromisos externos. Al asegurar que los compromisos reflejaran capacidad real en lugar de expectativas optimistas, la comunicación con el cliente transmitió una señal de confiabilidad operativa. La transparencia sobre las limitaciones, en lugar de gestionar las percepciones externamente, redujo la brecha entre lo que las partes clave esperaban y lo que la organización podía cumplir. Esto generó un movimiento positivo moderado en los indicadores de confianza de las partes clave."

["The decision to involve the operations director before confirming any timeline to the client established an internal validation process that protected the credibility of external commitments. By ensuring that commitments reflected actual capacity rather than optimistic expectations, the client communication conveyed a signal of operational reliability. The transparency about constraints, rather than managing perceptions externally, reduced the gap between what key parties expected and what the organization could deliver. This generated a moderate positive movement in key stakeholder confidence indicators."]

Gate checks: G1 ✓ G2 ✓ (mechanism: internal validation → credible external commitment → reduced expectation gap → trust) G3 ✓ G4 ✓ G5 ✓ (4 sentences) G6 ✓ **Signal anchoring note:** Component 1 echoes the student's stakeholder signal (operations director). Component 2 echoes justification signal (realistic capacity vs. optimistic expectations). Component 3 echoes ethical signal (transparency over perception management).

EXAMPLE D — PROHIBITED Explanation (What Not to Generate)

The following is shown as a negative example. It must never be generated.

"¡Excelente razonamiento! Tu decisión de consultar al equipo primero fue exactamente la acción correcta. La confianza de los stakeholders mejoró porque tomaste la mejor decisión posible. Si no

hubieras pensado en el equipo de operaciones, los resultados habrían sido mucho peores. Para la próxima decisión, sigue aplicando este tipo de pensamiento estratégico."

["Excellent reasoning! Your decision to consult the team first was exactly the right action. Stakeholder trust improved because you made the best possible decision. If you hadn't thought about the operations team, the results would have been much worse. For the next decision, keep applying this type of strategic thinking."]

Violations:

- *"¡Excelente razonamiento!"* — praise; evaluative
- *"exactamente la acción correcta"* — implies a correct answer
- *"la mejor decisión posible"* — superlative evaluation
- *"Si no hubieras pensado en..."* — counterfactual judgment
- *"Para la próxima decisión, sigue aplicando..."* — forward-looking hint
- No causal mechanism present — explanation jumps from decision to outcome without explaining how

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EXAMPLE E — Subtle Hint Failure (What Not to Generate)

This example passes surface-level checks but fails the Hint Test. It must not be generated.

"La confianza de los stakeholders disminuyó porque la decisión no incluyó una comunicación proactiva con las partes clave antes de ejecutar el cambio. Para evitar este tipo de impacto en decisiones futuras, es importante notificar a los stakeholders relevantes antes de implementar cambios que los afecten."

["Stakeholder trust declined because the decision did not include proactive communication with key parties before executing the change. To avoid this type of impact in future decisions, it is important to notify relevant stakeholders before implementing changes that affect them."]

Why it fails: The second sentence is a direct prescription for the next decision. It tells the student exactly what to do differently. This is answer-giving through the explanation channel.

Corrected version:

"La brecha entre la ejecución del cambio y la notificación de las partes con intereses directos en el resultado generó incertidumbre entre los stakeholders sobre la dirección de la organización. Esta incertidumbre afectó los indicadores de confianza en los stakeholders clave."

["The gap between the execution of the change and the notification of parties with direct interests in the outcome generated uncertainty among stakeholders about the organization's

direction. This uncertainty affected key stakeholder confidence indicators."]

5.12 Section Summary: What the Engine Must Do

At every decision turn, the causal explanation generator must:

1. **Generate one explanation per displayed KPI movement.** Every movement the student sees has an explanation ready before delivery. No movement is displayed without one.
2. **Apply the three-part structure** (Decision Reference, Causal Mechanism, Directional Connection) — compressed for Surface band, fully expressed for Engaged and Integrated.
3. **Scale depth to RDS band** — 2 sentences for Surface, 2–3 for Engaged, 3–4 for Integrated.
4. **Apply signal anchoring** at Engaged and Integrated bands — reflect the student's own reasoning back as organizational logic, without quoting directly or praising.
5. **Generate defensible MCQ explanations** from the pre-authored tradeoff signature for Turn 1.
6. **Generate defensible anti-pattern correction explanations** when Section 4 forced a movement — find the legitimate indirect causal pathway, never reveal the correction logic.
7. **Pass all six quality gates** — including the Hint Test — before delivering any explanation.
8. **Pass the complete explanation set to Section 6 (Response Assembly)** — one explanation per displayed KPI, tagged to the correct KPI ID and movement event.

Section 5 connects to: Section 2 (receives signal data and RDS band for anchoring and depth), Section 3 (receives narrative for consistency validation), Section 4 (receives KPI movement set — one explanation generated per displayed movement), Section 6 (passes complete explanation set for response assembly).